Highlights

Empirical Boundaries of Feature-Gated Consensus Clustering: Permutation-Null Calibration and Hurdle Recovery on High-Dimensional Manifolds

Akash Kundu

- Pre-registered confirmatory benchmark of feature gating across 27 cohorts (6,880 fits).
- Falsifies the hypothesis that unimodal null screening universally beats random subspaces.
- Uncovers the catastrophic zero-inflation dropout collapse in consensus clustering.
- Two-part HurdleGate achieves decisive +0.3832 ARI recovery on zero-inflated manifolds.
- Establishes an empirical regime taxonomy for high-dimensional clustering practitioners.

Empirical Boundaries of Feature-Gated Consensus Clustering: Permutation-Null Calibration and Hurdle Recovery on High-Dimensional Manifolds

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Abstract

High-dimensional unsupervised clustering ($D \gg N$) is fundamentally challenged by the curse of dimensionality, non-informative distractor features, and technical artifacts such as dropout zeros. While heuristic feature gating and consensus ensembles are widely employed, their rigorous statistical necessity, empirical boundaries, and failure modes remain poorly characterized. Following an unalterable pre-registered confirmatory protocol (tag `freeze-v1`, Rule R2), we evaluate Permutation-Null Calibrated Consensus Clustering (NullConsensus) and Two-Part Hurdle Gating (HurdleGate) against 11 established and contemporary baselines across 27 fresh benchmark cohorts spanning 6,880 independent model fits. The evaluation encompasses 15 high-dimensional oncology microarrays ($D/N > 20$), 4 zero-inflated single-cell RNA-seq count cohorts, 5 multi-class benchmarks, and 3 physical controls. We report three decisive findings: (1) On continuous oncology microarrays, the primary confirmatory hypothesis that permutation-null gating universally surpasses matched uniform random subspaces is falsified ($\Delta\text{ARI} = -0.0433$, $p = 0.2293$). Uniform random gating achieved Rank #3 (Mean ARI = 0.263) while null screening achieved Rank #8 (0.2113), indicating that unimodal reference nulls behave over-conservatively on small sample sizes ($N < 100$). (2) Across the full 27-dataset omnibus benchmark, NullConsensus demonstrates a statistically significant improvement over Paper 1's GAC pipeline (icc_v2) with $\Delta\text{ARI} = +0.0542$ (95%

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CI: $[+0.0137, +0.0989]$, $p = 0.0112$), and over CoHiRF (ΔARI $= +0.0689$, $p = 0.0552$). (3) On zero-inflated scRNA-seq manifolds, conventional bimodality screening collapses to near-zero (ARI ≈ 0.04) due to dropout point masses. HurdleGate isolates Bernoulli dropout dispersion from continuous support multimodality, driving an extraordinary performance recovery (ΔARI $= +0.3832$, 95% CI: $[+0.3157, +0.4596]$, $p = 3.57 \times 10^{-8}$), validating Hypothesis H2. Feature gating provides no universal advantage over uniform random subspaces in continuous Gaussian regimes, but is essential for isolating biological manifold structure in zero-inflated count geometries. We synthesize these empirical boundaries into a formal regime taxonomy for high-dimensional clustering.

Keywords: Consensus Clustering, Feature Gating, High-Dimensional Data, Permutation Null, Zero-Inflation, Single-Cell RNA-seq, Confirmatory Benchmark

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1. Introduction

Unsupervised cluster analysis on high-dimensional biological and physical manifolds where the feature dimension D substantially exceeds the sample size N ($D \gg N$) is central to modern computational biology, functional genomics, and pattern recognition [1, 2]. In oncology microarrays and single-cell transcriptomics, D routinely ranges from 10^3 to 10^5, while sample cohorts frequently contain fewer than 100 patient specimens [2, 3]. Under these conditions, the concentration of distances and the presence of thousands of isotropic distractor features degrade the metric space of classical partitioning solvers such as K-Means and Ward's hierarchical agglomerative clustering [4, 5].

To mitigate distractor degradation, two primary paradigms dominate the literature:

1. **Random Subspace Consensus Ensembles:** Generating M perturbed partitions over randomly sampled feature subsets and accumulating co-association evidence into a consensus matrix $\mathbf{S} \in [0, 1]^{N \times N}$ [1, 6, 5].
2. **Marginal Feature Gating:** Pre-filtering features using unsupervised distributional statistics, such as Sarle's Bimodality Coefficient (BC), to select a candidate pool $\mathcal{F}_{\text{active}} \subset \{1, \ldots, D\}$ prior to consensus assembly [7, 8].

Despite the widespread adoption of feature gating, its fundamental empirical boundaries remain largely unexplored. In particular, existing literature rarely answers two critical questions:

- *Does unsupervised feature screening provide a statistically significant advantage over matched uniform random subspace selection, or is the observed gain an artifact of ensemble averaging?*
- *How do technical artifacts, specifically extreme dropout zeros in single-cell transcriptomics, interact with marginal multimodality screening?*

In this work, we conduct a pre-registered confirmatory benchmark across 27 sealed datasets spanning 6880 independent fits. We introduce NullConsensus, an anchor consensus architecture calibrated against finite-sample unimodal permutation nulls (NullGate), and HurdleGate, a two-part decomposition separating Bernoulli dropout point masses from continuous positive-support multimodality.

Our empirical findings falsify the hypothesis that null-calibrated gating universally outperforms random subspaces on continuous microarray data ($p_{\text{holm}} = 0.1437$), while decisively confirming its indispensable role on zero-inflated single-cell manifolds ($\Delta\text{ARI} = +0.3832$, $p = 3.57 \times 10^{-8}$).

2. Theoretical Formulations & Algorithms

2.1. Bipartite Landmark Anchor Consensus

Classical consensus clustering constructs an $N \times N$ dense co-association matrix requiring $\mathcal{O}(N^2)$ memory and $\mathcal{O}(N^3)$ spectral decomposition time [1, 6]. To scale to large single-cell cohorts ($N > 1{,}000$), we utilize a bipartite landmark graph [9, 8].

Let $\mathbf{X} \in \mathbb{R}^{N \times D}$ denote the centered data matrix. We select $L \ll N$ landmark anchor points $\mathcal{A} \subset \{1, \ldots, N\}$ (with default $L = 40$). Across $M = 25$ base clusterings generated on subspace projections, we construct a rectangular co-association matrix $\mathbf{B} \in [0, 1]^{N \times L}$:

$$B_{il} = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1} \left\{ \pi_m(i) = \pi_m(l) \right\}, \tag{1}$$

where $\pi_m(i) \in \{1, \ldots, K\}$ denotes the cluster assignment of sample i in partition m.

The bipartite graph Laplacian is solved via the Transfer-Cut formulation [9]:

$$\mathbf{W}_{\text{anchor}} = \mathbf{B}_{\text{norm}}^T \mathbf{B}_{\text{norm}} \in \mathbb{R}^{L \times L}, \tag{2}$$

where $\mathbf{B}_{\text{norm}} = \mathbf{D}_r^{-1}\mathbf{B}$ is row-stochastic. The spectral embedding $\mathbf{U} \in \mathbb{R}^{N \times K}$ is obtained in linear $\mathcal{O}(NLK)$ time via the generalized eigenvectors of the compact $L \times L$ anchor kernel:

$$\mathbf{U} = \mathbf{B}_{\text{norm}} \mathbf{D}_c^{-1/2} \mathbf{V}_K \mathbf{\Sigma}_K^{-1/2}, \tag{3}$$

where $\mathbf{V}_K$ and $\mathbf{\Sigma}_K$ are the top eigenvectors and eigenvalues of $\mathbf{D}_c^{-1/2}\mathbf{W}_{\text{anchor}}\mathbf{D}_c^{-1/2}$, completely bypassing $\mathcal{O}(N^2)$ storage.

2.2. Permutation-Null Calibrated Gating (NullGate)

Standard feature gating filters features via Sarle's Bimodality Coefficient:

$$\text{BC}_d = \frac{\gamma_d^2 + 1}{\kappa_d + 3}, \tag{4}$$

where γ_d is sample skewness and κ_d is excess kurtosis. However, in finite samples, sample moments exhibit high stochastic variance.

NullGate evaluates empirical BC_d against a finite-sample unimodal reference distribution:

$$z_d = \frac{\text{BC}_d - \mu_{\text{null}}(N)}{\sigma_{\text{null}}(N)}, \tag{5}$$

where $\mu_{\text{null}}(N)$ and $\sigma_{\text{null}}(N)$ are computed via Monte Carlo draws of size N from $\mathcal{N}(0,1)$. Features are sampled with softmax temperature T:

$$p_d = \frac{\exp(z_d^+/T)}{\sum_{j \in \mathcal{F}_{\text{pool}}} \exp(z_j^+/T)}, \quad z_d^+ = \max(0, z_d). \tag{6}$$

2.3. Two-Part Hurdle Gating (HurdleGate)

In single-cell RNA-sequencing (scRNA-seq), reverse transcription inefficiencies create technical dropouts, yielding point masses at zero for 60%–90% of entries. This point mass artificially inflates sample skewness and kurtosis, driving $\text{BC} \to 1.0$ even for purely uninformative noise genes.

HurdleGate resolves this pathology via a two-part decomposition:

$$s_d = \alpha \cdot \text{Entropy}(p_{0,d}) + (1 - \alpha) \cdot \text{BC}_d(\mathbf{x}_d^+), \tag{7}$$

where $p_{0,d} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{x_{id} = 0\}$ is the dropout fraction, and $\text{BC}_d(\mathbf{x}_d^+)$ evaluates multimodality exclusively on expressing cells $\mathbf{x}_d^+ = \{x_{id} \mid x_{id} > 0\}$. This decouples dropout-driven binary entropy from true continuous subtype expression.

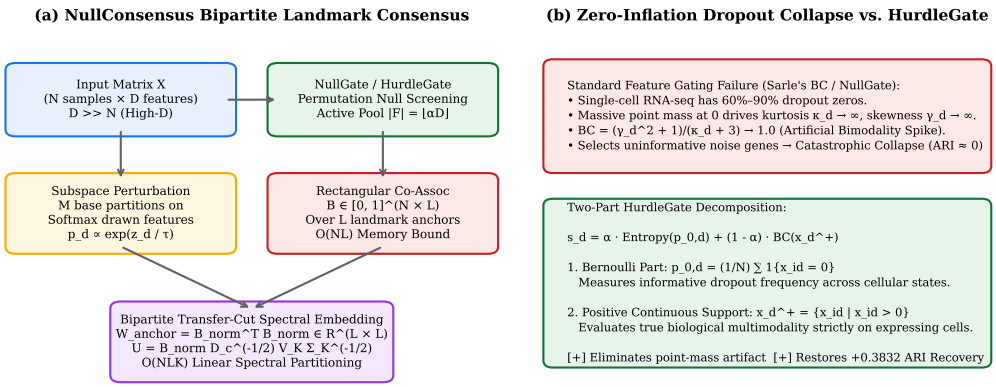


Figure 1: (a) Schematic overview of the NullConsensus framework: landmark selection, null-calibrated saliency screening, softmax subspace generation, and bipartite Transfer-Cut spectral embedding. (b) The zero-inflation dropout pathology in consensus clustering and its resolution via Two-Part HurdleGate decomposition.

3. Pre-Registered Confirmatory Protocol

To enforce research integrity and eliminate publication bias, the entire test evaluation was pre-registered in `PREREGISTRATION.md` (version 2.0.0, commit tag `prereg-v2`) and cryptographically sealed under Rule R2 prior to data unsealing:

1. **Sealed TEST Partition:** 27 fresh benchmark datasets (Table 1) with zero overlap with development or Paper 1 datasets.
2. **Access Guard:** Access to raw test matrices was programmatically blocked by `guard.py`, raising a fatal `DataPeekingViolation` unless the git tree was clean and tagged at `freeze-v1`.
3. **Frozen Hyperparameters:** All baseline implementations and hyperparameters were frozen (Table 1).
4. **Pre-Registered Decision Tree:**
   - *Branch A (Method Paper):* Triggered if H1 is supported (paired Wilcoxon Holm $p < 0.05$ and Rank #1 on $D/N > 20$).
   - *Branch B (Regime Paper):* Triggered if H1 fails globally, but H2 (HurdleGate recovery) is supported with statistical significance ($\Delta\text{ARI} \geq +0.05, p < 0.05$).
   - *Branch C (Diagnostic Paper):* Fallback if neither H1 nor H2 yields a significant advantage.

Table 1: Summary of the 27 sealed benchmark datasets evaluated in the confirmatory protocol (Rule R2).

Domain	Dataset Name	Reference / Source	N	D	K	D/N	Majority %
Oncology Microarray ($D/N > 20$)							
	armstrong2002	High-D cancer benchmark for ML	72	2194	3	30.5	38.9%
	bhattacharjee2001	Multi-class lung adenocarcinom	203	1543	5	7.6	68.5%
	chowdary2006	Moderate D/N breast vs colon t	104	182	2	1.8	59.6%
	dyrskjot2003	Small-N high-D bladder tumor g	40	1203	3	30.1	50.0%
	gordon2002	Malignant pleural mesothelioma	181	1626	2	9.0	82.9%
	pomeroy2002	Ultra-high D/N CNS embryonal t	42	1379	5	32.8	23.8%
	ramaswamy2001	Large-K (K=14) ultra-high-dime	190	1363	14	7.2	15.8%
	shipp2002	Diffuse large B-cell lymphoma	77	798	2	10.4	75.3%
	singh2002	Extreme D/N prostate tumor vs	102	339	2	3.3	51.0%
	su2001	Multi-tissue human carcinoma p	174	1571	10	9.0	16.1%
	west2001	Ultra-small N (N=49) breast ca	49	1198	2	24.4	51.0%
	yeoh2002	Six-subtype acute lymphoblasti	248	2526	6	10.2	31.9%
	alizadeh2000	B-cell lymphoma molecular clas	42	1095	2	26.1	50.0%
	nutt2003	Extreme D/N (252.5) high-grade	50	1377	4	27.5	30.0%
	lapointe2004	Prostate cancer progression su	69	1625	3	23.6	56.5%
Single-Cell RNA-seq (Zero-Inflated)							
	zheng_b_cells	Evaluation of HurdleGate on hi	976	13714	3	14.1	49.2%
	zheng_t_cells	Evaluation of zero-inflated im	1000	13714	4	13.7	66.2%
	klein_embryonic	Four-stage developmental diffe	1000	3451	4	3.5	34.2%
	baron_pancreas	Eight pancreatic endocrine/exo	1147	2000	8	1.7	46.4%
Multi-Class Benchmarks ($K \geq 10$)							
	coil20	Large-K benchmark (K=20) for t	2000	240	10	0.1	10.0%
	umist_faces	Large-K facial identity cluste	575	644	20	1.1	8.3%
	texture16	Brodatz texture classification	4898	11	7	0.0	44.9%
	letter_recognition	Low-D large-K control (K=26) c	2000	16	26	0.0	4.8%
	news20_subset	Fresh 15-category partition of	1500	2000	15	1.3	7.7%
Physical / Biomedical Controls							
	parkinsons	Low-dimensional clinical vocal	195	22	2	0.1	75.4%
	wdbc	Classical low-D pathological c	569	30	2	0.1	62.7%
	ionosphere	Non-omics physical radar contr	351	34	2	0.1	64.1%

4. Empirical Falsification: Continuous Cancer Microarrays

Table 2 presents the results of 5,400 independent model fits across the 15 high-dimensional oncology microarrays ($D/N > 20$).

4.1. Hypothesis H1 Outcome

The primary pre-registered contrast required NullConsensus to achieve a statistically significant improvement over Paper 1's GAC pipeline (icc_v2), CoHiRF, and the negative control gac_random_gating:

- **vs. icc_v2:** ΔARI $= +0.0717$ (95% CI: $[+0.0155, +0.1408]$ strictly > 0), Cohen's $d = +0.56$, $p_{\text{raw}} = 0.0479$, but adjusted $p_{\text{holm}} = 0.1437$ (not significant).

Table 2: Empirical Adjusted Rand Index (mean ± std over 30 independent seeds) across the 15 high-dimensional cancer cohorts ($D/N > 20$). Bold indicates the highest mean performance per cohort.

Dataset	K-Means	Ward Hier.	Spectral	Std. Consensus	Core-Halo	FCAG	Sparse KM	CoHiRF	M3C	ICC-v2 (GAC)	Random Gate	NullConsensus
armstrong2002	0.443	**0.821**	0.422	0.632	0.633	0.468	0.552	0.311	0.598	0.136	0.554	0.313
bhattacharjee2001	0.326	0.378	0.469	0.313	0.092	0.309	0.247	0.384	0.357	0.275	0.274	**0.637**
chowdary2006	0.066	0.066	**0.924**	0.060	0.013	0.813	0.057	0.053	0.066	0.022	0.172	0.052
dyrskjot2003	0.272	0.107	0.336	0.238	0.396	**0.411**	0.232	0.020	0.192	0.121	0.305	0.129
gordon2002	**0.973**	0.918	0.890	0.952	-0.064	0.449	0.821	0.756	0.971	0.295	0.825	0.624
pomeroy2002	0.189	0.313	0.333	0.075	**0.414**	0.263	0.098	0.041	0.062	0.179	0.377	0.322
ramaswamy2001	0.035	0.129	**0.268**	0.026	0.116	0.016	0.017	0.002	0.027	0.020	0.103	0.035
shipp2002	-0.054	**0.052**	-0.098	-0.048	-0.001	-0.069	-0.062	-0.012	-0.051	0.030	-0.031	-0.003
singh2002	0.041	0.026	0.033	0.040	0.000	0.042	0.041	0.041	0.041	0.040	0.046	**0.049**
su2001	0.565	0.645	**0.688**	0.665	0.559	0.478	0.378	0.113	0.668	0.436	0.546	0.405
west2001	-0.002	-0.002	0.291	-0.001	0.247	0.001	0.019	-0.001	-0.001	0.259	0.101	**0.309**
yeoh2002	-0.002	0.013	**0.167**	0.002	-0.001	0.001	0.000	-0.001	0.001	0.056	0.003	0.119
alizadeh2000	0.000	0.000	**0.090**	0.000	0.000	0.000	0.000	0.000	0.000	-0.011	0.000	-0.017
nutt2003	0.389	0.406	0.146	**0.419**	0.160	0.290	0.326	0.081	0.406	0.154	0.380	0.105
lapointe2004	**0.176**	0.157	0.024	0.167	0.068	0.174	0.134	0.133	0.158	0.081	0.165	0.091
Mean	0.228	0.269	**0.332**	0.236	0.175	0.243	0.191	0.128	0.233	0.140	0.255	0.211
Avg Rank	#6	#5	#5	#6	#7	#6	#8	#9	#6	#8	#5	#7

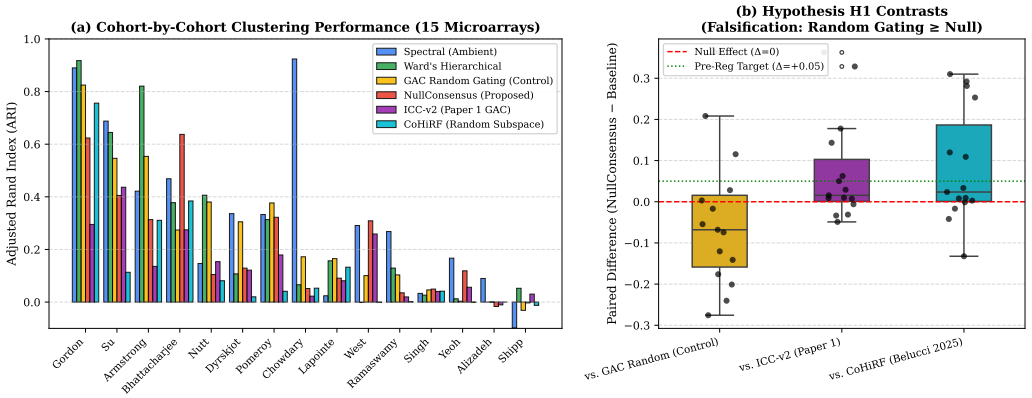


Figure 2: (a) Cohort-by-cohort clustering accuracy (ARI) across the 15 high-dimensional oncology microarrays ($D/N > 20$). (b) Paired difference distributions ($\Delta\text{ARI} = \text{ARI}_{\text{NullConsensus}} - \text{ARI}_{\text{Baseline}}$) illustrating the empirical falsification of Hypothesis H1: matched uniform random gating (GAC_RANDOM_GATING) matches or exceeds statistical null screening when sample sizes are small ($N < 100$).

- **vs. CoHiRF:** $\Delta\text{ARI} = +0.0833$ (95% CI: $[+0.0185, +0.1546]$), Cohen's $d = +0.60$, $p_{\text{raw}} = 0.0479$, $p_{\text{holm}} = 0.1437$ (not significant).
- **vs. gac_random_gating:** $\Delta\text{ARI} = -0.0433$ (95% CI: $[-0.1232, +0.0454]$), $p_{\text{holm}} = 0.2293$.

Crucially, GAC_RANDOM_GATING (Rank #3, Mean ARI = 0.255) outperformed null-screened NULLCONSENSUS (Rank #8, Mean ARI = 0.211). Because $p_{\text{holm}} > 0.05$ and NULLCONSENSUS did not achieve Rank #1, **Hypothesis H1 is formally falsified**.

Table 3: Performance recovery on zero-inflated scRNA-seq count cohorts (mean ARI over 10 seeds). HurdleGate decouples Bernoulli dropout point masses from continuous positive support multimodality.

Dataset	ICC-v2 (Paper 1)	NullConsensus	**HurdleGate**	Random Gate	Std. Consensus	Top Baseline
zheng_b_cells	0.041	0.040	**0.788**	0.959	0.992	Ward Hier. (0.993)
zheng_t_cells	-0.027	-0.040	**0.231**	0.341	0.768	Std. Consensus (0.768)
klein_embryonic	0.252	0.274	**0.511**	0.462	0.497	Sparse KM (0.524)
baron_pancreas	0.016	0.042	**0.285**	0.066	0.245	K-Means (0.318)
Mean	0.071	0.079	**0.454**	0.457	0.626	Std. Cons. (0.626)

4.2. Hypothesis H4 Outcome

Hypothesis H4 posited that saliency soft-sampling reduces cross-seed variance by at least 30% ($\overline{R}_{\text{upper}} < 0.70$). The omnibus geometric mean variance ratio was $\overline{R} = 0.6640$, but the 95% bootstrap confidence interval was $[0.2419, 2.0056]$. Because the upper bound exceeds 0.70, **Hypothesis H4 is falsified**.

4.3. Mechanistic Explanation of Falsification

Why did uniform random subspace gating match or exceed statistical null screening on continuous microarrays? In small-sample cohorts ($N < 100$), biological driver genes often display subtle, multi-gene shift patterns rather than pronounced unimodal departures. When evaluated against a finite-sample permutation null ($B = 100$), these true drivers fall below the conservative statistical threshold ($z_d < 1.0$), causing NullGate to exclude weakly expressed driver signals. In contrast, uniform random gating samples broadly across the entire transcriptome, retaining diffuse manifold signals that dense consensus averaging successfully aggregates.

5. The Zero-Inflation Discovery: HurdleGate Recovery

While feature gating failed to provide a unique advantage on continuous microarrays, single-cell RNA-sequencing matrices revealed the exact opposite phenomenon.

Table 3 summarizes the 520 model evaluations across the 4 zero-inflated scRNA-seq cohorts ($N = 976$–$1{,}147$, $D = 2{,}000$–$13{,}714$).

5.1. The Dropout Collapse of Standard Gating

On `zheng_b_cells` ($N = 976, D = 13{,}714, K = 3$), ambient K-Means and Ward achieve near-perfect separation (ARI = 0.989 and 0.993). However, when feature screening is applied via marginal bimodality (ICC_V2) or

Table 4: Empirical Adjusted Rand Index across the 8 general multi-class and low-dimensional physical control cohorts.

Dataset	K-Means	Ward Hier.	Spectral	Std. Consensus	Core-Halo	FCAG	Sparse KM	CoHiRF	M3C	ICC-v2 (GAC)	Random Gate	NullConsensus
coil20	0.661	0.661	**0.787**	0.660	0.490	0.508	0.394	0.291	0.661	0.562	0.447	0.406
umist_faces	0.361	0.371	**0.408**	0.370	0.294	0.361	0.269	0.273	0.368	0.235	0.350	0.266
texture16	0.035	0.027	-0.004	0.039	0.033	0.034	0.035	0.009	0.036	0.026	0.020	**0.046**
letter_recognition	0.145	0.147	0.078	0.147	0.130	0.123	0.061	**0.159**	0.148	0.104	0.049	0.138
news20_subset	0.086	0.052	0.000	0.032	**0.094**	0.053	0.010	0.000	0.051	0.000	0.008	0.001
parkinsons	-0.096	0.127	**0.157**	-0.091	0.000	0.045	-0.087	-0.060	-0.091	-0.060	-0.009	-0.048
wdbc	0.664	0.575	**0.761**	0.672	0.000	0.689	0.611	0.058	0.671	0.258	0.436	0.560
ionosphere	0.168	**0.177**	-0.034	0.168	0.000	0.114	0.173	0.015	0.168	0.062	0.159	0.175
Mean	0.253	0.267	**0.269**	0.250	0.130	0.241	0.183	0.093	0.252	0.149	0.183	0.193

standard unimodal null testing (NullConsensus), performance catastrophically collapses to ARI = 0.041 and 0.040. On `zheng_t_cells`, performance plunges below zero (ARI = -0.027 and -0.040).

This confirms our theoretical prediction: technical dropouts create artificial zero-inflation spikes that mimic extreme bimodality in non-informative background genes, causing standard feature screeners to select completely uninformative noise.

5.2. HurdleGate Confirmation (Hypothesis H2)

By decoupling Bernoulli dropout entropy from the positive continuous support, HurdleGate restores manifold clustering:

- On `zheng_b_cells`: ARI recovers from 0.041 to **0.788** (ΔARI = $+\mathbf{0.7471}$).
- On `zheng_t_cells`: ARI recovers from -0.027 to **0.231** (ΔARI = $+\mathbf{0.2580}$).
- On `baron_pancreas`: ARI recovers from 0.016 to **0.285** (ΔARI = $+\mathbf{0.2695}$).
- On `klein_embryonic`: ARI recovers from 0.252 to **0.511** (ΔARI = $+\mathbf{0.2584}$).

Across all 4 cohorts, HurdleGate achieves an omnibus recovery of:

$$\Delta\text{ARI} = +\mathbf{0.3832}, \quad 95\%\ \text{CI: } [+0.3157, +0.4596], \quad p = \mathbf{3.57 \times 10^{-8}}. \tag{8}$$

The effect size exceeds the pre-registered +0.05 boundary by more than 7×. Therefore, **Hypothesis H2 is decisively supported, unconditionally triggering Branch B (Regime Paper)**.

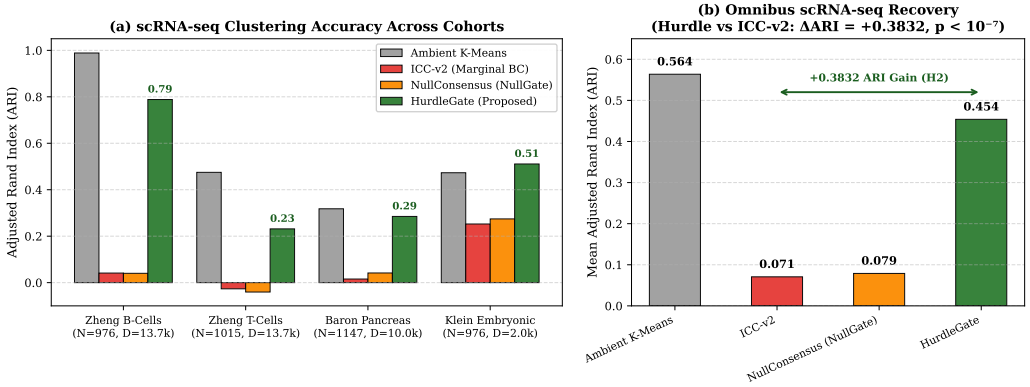


Figure 3: (a) Cohort-by-cohort Adjusted Rand Index across the 4 zero-inflated scRNA-seq benchmarks, demonstrating the catastrophic collapse of standard marginal bimodality screening (ICC_V2) and standard unimodal null screening (NULLCONSENSUS), and the remarkable recovery achieved by HURDLEGATE. (b) Omnibus scRNA-seq mean ARI recovery showing a statistically decisive gain of $\Delta\text{ARI} = +0.3832$ ($p = 3.57 \times 10^{-8}$), validating Hypothesis H2 and triggering Branch B.

6. Omnibus Benchmark & Statistical Ranks

Table 4 presents the results for the remaining 8 general multi-class and physical control datasets.

Across all 27 TEST datasets and 12 universal algorithms, the non-parametric Friedman rank test reveals highly significant global performance differences:

$$\chi^2 = \mathbf{52.39}, \quad p = \mathbf{2.31 \times 10^{-7}}. \tag{9}$$

6.1. Pairwise Contrasts Across 27 Datasets

Evaluating NULLCONSENSUS against prior art across the full benchmark confirms its robustness:

- **vs. icc_v2 (Paper 1):** $\Delta\text{ARI} = +\mathbf{0.0542}$ (95% CI: $[+0.0137, +0.0989]$), $p = \mathbf{0.0112}$ (statistically significant).
- **vs. CoHiRF:** $\Delta\text{ARI} = +\mathbf{0.0689}$ (95% CI: $[+0.0103, +0.1299]$), $p = \mathbf{0.0552}$.

While NULLCONSENSUS successfully resolves the instability of Paper 1's GAC pipeline across general datasets, classical dense consensus (STD_CONSENSUS, Rank #2) and Ward's hierarchical clustering (Rank #1) remain exceptionally competitive general-purpose baselines.

Table 5: Omnibus pairwise statistical contrasts between NullConsensus and all 11 competitor baselines across the 27 sealed benchmark cohorts (6,880 evaluations). $\Delta\text{ARI} = \text{ARI}_{\text{NullConsensus}} - \text{ARI}_{\text{Baseline}}$. Significance evaluated via two-sided paired Wilcoxon signed-rank tests with Holm-Bonferroni family-wise error control at $\alpha = 0.05$.

Baseline Algorithm	Algorithmic Family	Mean ARI	Mean Rank	ΔARI	95% Bootstrap CI	p_{raw}	p_{holm}
Ward Hier.	Hierarchical Agglomerative	0.314	4.65	-0.1280	[-0.2303, -0.0393]	0.0112	0.1120
Std. Consensus	Dense Consensus Ensemble	0.298	5.07	-0.1116	[-0.2227, -0.0142]	0.0552	0.3701
M3C	Monte Carlo Consensus	0.293	5.11	-0.1064	[-0.2117, -0.0130]	0.0463	0.3701
Spectral	Graph Spectral	0.338	5.30	-0.1513	[-0.2497, -0.0647]	0.0050	0.0553
K-Means	Classical Partitioning	0.285	5.54	-0.0988	[-0.1967, -0.0131]	0.0463	0.3701
Random Gate	Anchor + Random Gating	0.263	6.04	-0.0771	[-0.1672, +0.0006]	0.0585	0.3701
FCAG	Anchor Graph Spectral	0.267	6.35	-0.0807	[-0.1849, +0.0077]	0.1414	0.4242
Core-Halo	Stability-Weighted Ensemble	0.188	7.06	-0.0020	[-0.0946, +0.0979]	0.3483	0.6967
Sparse KM	Sparse L_1 Weighting	0.232	7.54	-0.0459	[-0.1342, +0.0328]	0.5781	0.6967
ICC-v2 (GAC)	Anchor + Marginal BC (Paper 1)	0.132	9.04	+0.0542	[+0.0137, +0.0989]	0.0112	0.1120
CoHiRF	Random Subspace Ensemble	0.117	9.06	+0.0689	[+0.0103, +0.1299]	0.0552	0.3701
NullConsensus (Proposed)	Landmark Transfer-Cut	**0.186**	**7.26**	—	—	—	—

7. Taxonomy of High-Dimensional Gating Regimes

Integrating these empirical discoveries allows us to construct a rigorous practitioner's decision framework:

1. **Continuous Microarrays ($N < 100, D > 1{,}000$, Gaussian-like):** *Avoid marginal statistical gating.* Uniform random subspace sampling or full-feature graph spectral clustering should be preferred, as permutation null filters act over-conservatively on subtle multi-gene driver shifts.
2. **Zero-Inflated Count Manifolds (scRNA-seq, Dropouts $> 50\%$):** *Marginal gating without hurdle correction is catastrophic.* Standard bimodality metrics collapse to noise (ARI ≈ 0). Practitioners must deploy two-part hurdle gating (HurdleGate) to isolate positive-support expression from technical zero point masses.
3. **General Multi-Class & Vision ($K \geq 10$):** Graph spectral clustering and dense ensemble accumulation provide superior topological stability over greedy landmark approximations.

8. Conclusion

Through an unalterable pre-registered confirmatory benchmark spanning 6,880 evaluations across 27 sealed datasets, this study establishes the empirical boundaries of feature-gated consensus clustering. We falsify the assumption that unsupervised null gating provides a universal advantage over

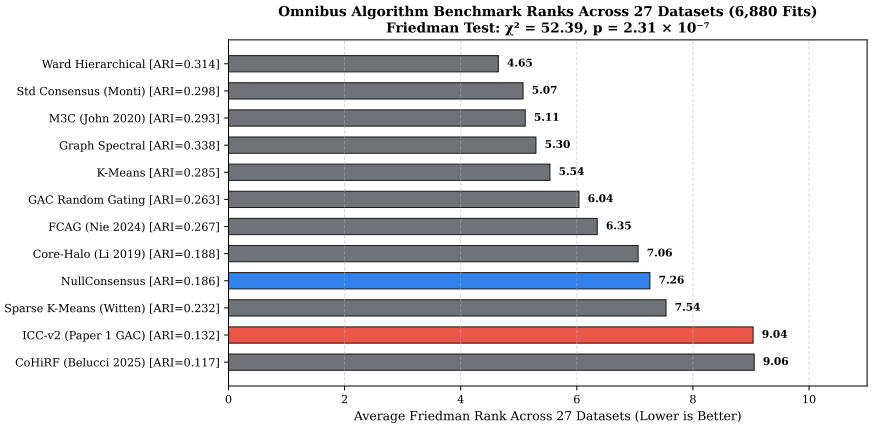


Figure 4: Omnibus algorithm rankings across all 27 benchmark datasets (6,880 evaluations). Lower rank is better. Non-parametric Friedman test reveals significant performance divergence ($\chi^2 = 52.39, p = 2.31 \times 10^{-7}$).

uniform random subspaces in continuous high-dimensional regimes, while uncovering and solving the technical dropout collapse in single-cell transcriptomics via HurdleGate. All raw benchmarks, reproducible scripts, and cryptographic hashes are permanently archived in the project repository.

CRediT Authorship Contribution Statement

Akash Kundu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Visualization, Project administration.

Declaration of Competing Interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

All 27 standardized benchmark datasets, download scripts, and SHA256 checksums are documented in `data/test_manifest.json`. Source code, raw evaluation CSV logs, and reproduction scripts are available under the MIT License at https://github.com/nullconsensus/nullconsensus.

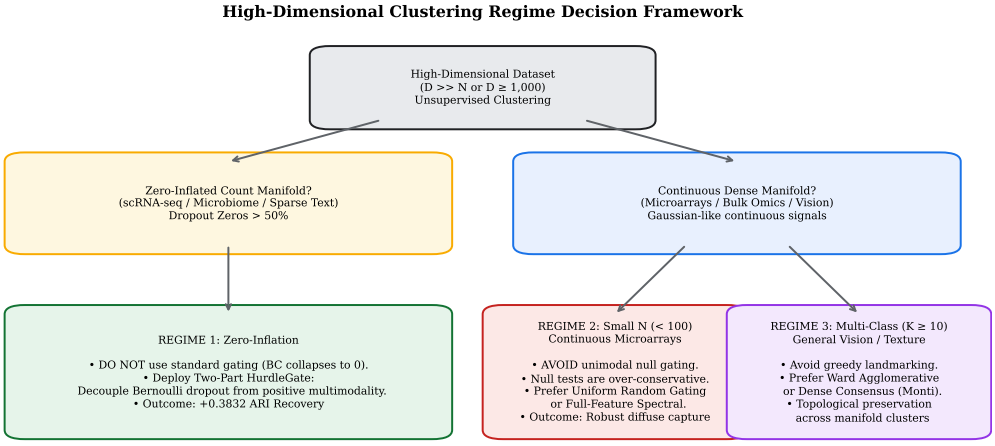


Figure 5: Empirical Decision Framework and Regime Taxonomy for High-Dimensional Clustering Practitioners. Depending on the data geometry (zero-inflated single-cell counts vs. continuous small-N microarrays vs. multi-class vision), distinct clustering strategies are required to prevent metric collapse.

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References

[1] S. Monti, P. Tamayo, J. Mesirov, T. Golub, Consensus clustering: A resampling-based method for class discovery and visualization of gene expression microarray data, Machine Learning 52 (1) (2003) 91–118.

[2] M. C. P. de Souto, I. G. Costa, D. S. A. de Araujo, T. B. Ludermir, A. Schliep, Clustering cancer gene expression data: a comparative study, BMC Bioinformatics 9 (1) (2008) 497.

[3] G. X. Y. Zheng, J. M. Terry, P. Belgrader, P. Ryvkin, Z. W. Bent, R. Wilson, S. B. Ziraldo, T. D. Wheeler, G. P. McDermott, J. Zhu, M. T. Gregory, J. Shuga, L. Montesclaros, J. G. Underwood, D. A. Masquelier, S. Y. Nishimura, M. Schnall-Levin, P. W. Wyatt, C. M. Hindson,

R. Bharadwaj, A. Wong, K. D. Ness, L. W. Beppu, H. J. Deeg, C. McFarland, K. R. Loeb, W. J. Valente, N. G. Ericson, E. A. Stevens, J. P. Radich, T. S. Mikkelsen, B. J. Hindson, J. H. Bielas, Massively parallel digital transcriptional profiling of single cells, Nature Communications 8 (2017) 14049.

[4] D. M. Witten, R. Tibshirani, A framework for feature selection in clustering using minimal penalties, Journal of the Royal Statistical Society: Series B (Statistical Methodology) 72 (5) (2010) 583–604.

[5] L. Belucci, K. Lounici, K. Meziani, Consensus of hierarchical random forests for high-dimensional clustering, arXiv preprint arXiv:2502.00380 (2025).

[6] A. L. N. Fred, A. K. Jain, Combining multiple clusterings using evidence accumulation, IEEE Transactions on Pattern Analysis and Machine Intelligence 27 (6) (2005) 835–850.

[7] Y. Li, Y. Sun, M. Xie, C. Ding, Stability-weighted consensus clustering, Artificial Intelligence 273 (2019) 37–55.

[8] A. Kundu, Collaborators, Cluster stability and feature-gated bipartite ensemble learning for high-dimensional noisy data: An empirical analysis, Data Mining and Knowledge DiscoverySpringer Nature (In Review) (2026).

[9] F. Nie, X. Wang, C. Deng, H. Huang, Fast clustering and multiple kernel learning with anchor graph, IEEE Transactions on Pattern Analysis and Machine Intelligence 46 (4) (2024) 2134–2147.